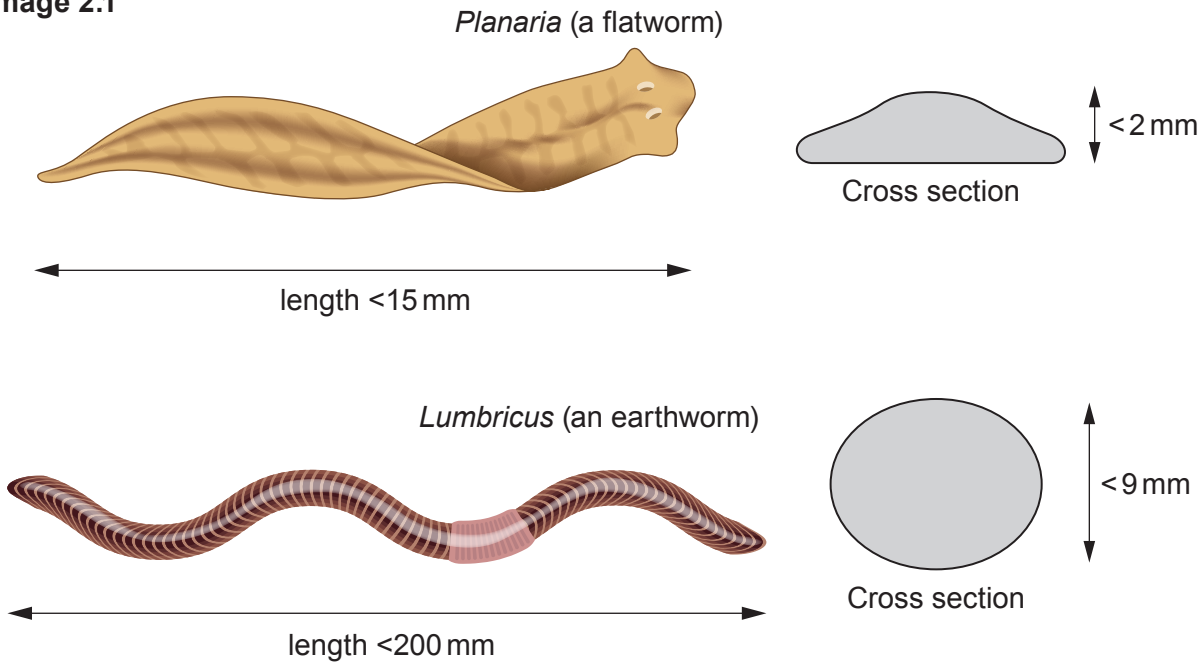


2. *Planaria* and *Lumbricus* are both multicellular animals that use their body surface for gas exchange. These are shown in **Image 2.1**.

Image 2.1



(a) (i) The body surface of both animals is adapted for gas exchange by being thin, moist, and permeable to gases. State **one** other feature of the body surface that these organisms have in common. [1]

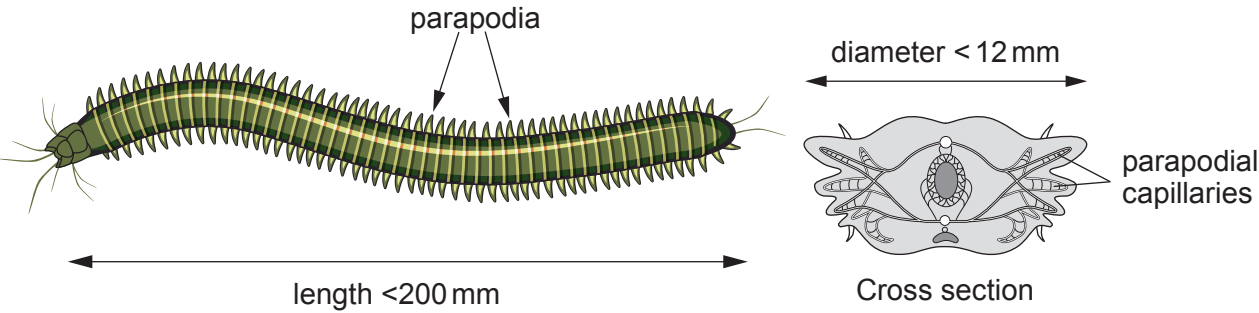
(ii) *Planaria* does not require a circulatory system. Explain why *Lumbricus* does require a circulatory system. [3]

(iii) Locusts are large insects. Explain how they ensure an efficient supply of oxygen to their tissues. [2]



(b) The animal shown in **Image 2.2** is *Nereis* (a ragworm).

Image 2.2



Nereis also uses its body surface for gas exchange and, like *Lumbricus*, has a closed circulatory system and blood containing haemoglobin. *Lumbricus* is slow-moving and burrows in damp soil feeding on decaying organic matter. *Nereis* is a fast-moving, marine predator and uses its parapodia to crawl and swim.

- (i) Suggest **how** parapodia are also adapted to increase the efficiency of gas exchange.

[1]

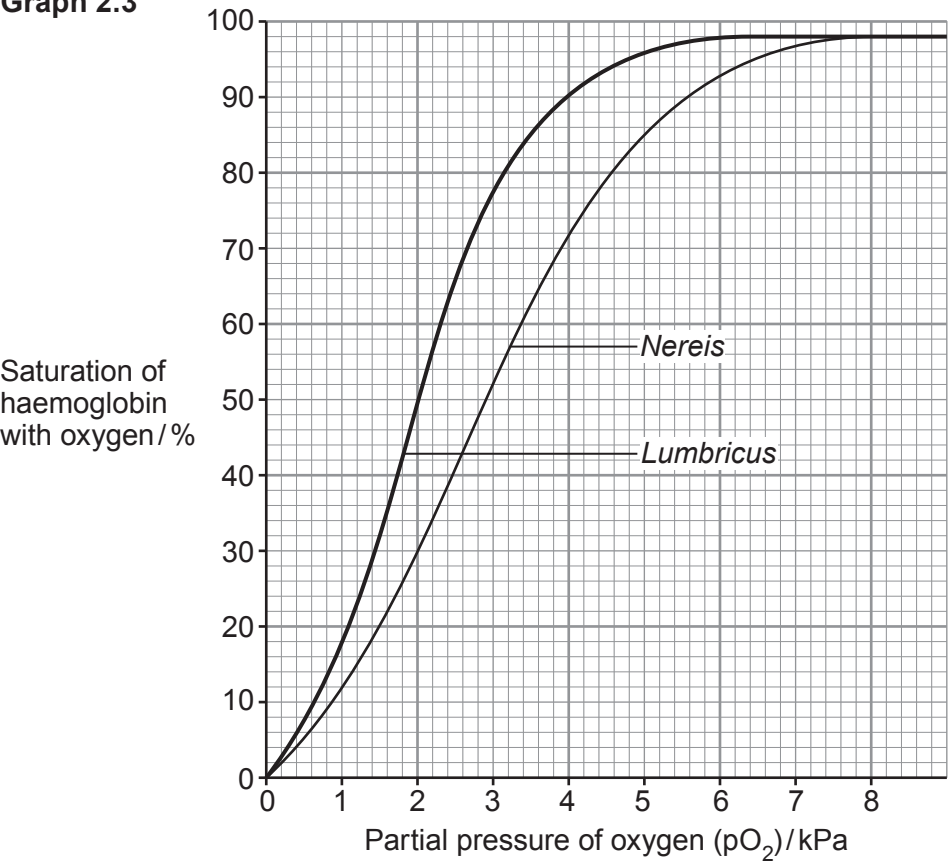
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Graph 2.3 shows the oxygen haemoglobin dissociation curves for *Lumbricus* and *Nereis* haemoglobin.

Graph 2.3



- (ii) Use the information provided to explain how the difference in the positions of the dissociation curves for *Lumbricus* and *Nereis* haemoglobin reflects their method of feeding. [5]

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